P vs NP: Finite-Grain Blindness, Completion, and Canonical Realization

Fabrizio De Palma
Independent Researcher

Abstract

We introduce a topology on the space

$$X = \mathcal{P}(\{0,1\}^*)$$

of languages, induced by a finite-grain filtration of computational indistinguishability. In this topology, P and NP form open regions, and the computational difficulty of separating them is localized on the boundary $\partial P \cap NP$. We prove that every monotone separator determined at a finite stage of the filtration is locally constant and therefore blind on the P–NP boundary. This yields a structural no-go theorem for finite-grain approaches **[1, 2, 8]**.

We then study the inverse-limit completion

$$\hat{X} = \lim_{k} \leftarrow \; X/\sim_k$$

of the filtration. Inside $\hat{X}$, we define realization towers, canonical realizations, and persistent frontier points. We prove that internal persistent frontier points which are non-rigid admit no canonical realization. As a consequence, passing from finite-grain quotients to the completion does not automatically restore canonical separability: the finite-level obstruction may persist at the limit level as failure of canonicity **[4, 7]**.

The paper does not resolve Pvs NP. Its contribution is structural: it isolates two distinct obstructions — finite-grain blindness and completion-level non-canonicity — and shows that any successful separator must overcome both.

Keywords: P vs NP, Computational Complexity, Complexity Theory, Polynomial-Time Reductions, Topological Methods, Structural Barriers, Inverse Limits, Finite-Grain Blindness

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1. Introduction

The problem of separating the complexity classes P and NP goes back to the foundational works of Cook and Karp **[3, 6]**. Classical barriers such as relativization, natural proofs, and algebrization already show that broad classes of methods cannot resolve the problem **[1, 2, 8]**.

Let

$$X = \mathcal{P}(\{0,1\}^*)$$

be the space of languages. The purpose of this paper is to study the P vs NP problem through a filtration of finite-grain computational indistinguishability on X. The basic idea is that bounded structural observation induces a topology on X, and that the resulting geometry already constrains broad classes of possible separators [**5**].

The first part of the paper is finite-level. Given a filtration

$$(\sim_k)_{k\in\mathbb{N}}$$

we define neighborhoods

$$N_k(A) = [A]_{\sim_k}$$

and the induced topology of finite observability. In this topology, reducibility-stable classes are open; in particular, P and NP are open subsets of X. The relevant region is therefore not the interior of either class, but the boundary

$$\partial P \cap NP.$$

We prove that every monotone invariant determined at some finite level k is constant on $N_k(A)$. Hence every such invariant is blind on the computational boundary. This is the first main result: **finite-grain monotone methods cannot separate P from NP where the problem actually lives**.

The second part of the paper addresses the natural objection to this conclusion. If no finite stage of the filtration suffices, one should pass to the inverse-limit object associated with the entire tower. We therefore introduce the completion [**4, 7**]

$$\hat{X} = \lim_{k} \leftarrow \; X/\sim_k.$$

A point of $\hat{X}$ is a coherent profile of finite-grain equivalence classes. The mathematical question is then no longer whether finite information is sufficient, but whether the resulting inverse-limit data admit a **canonical realization**.

To study this question, we define **realization towers**, **internal canonical selections**, and **persistent frontier points** in $\hat{X}$. The key distinction is between existence and canonicity: a point of the completion may admit coherent realizations without determining a unique structurally privileged branch. Our second main result states that **internal persistent frontier points which are non-rigid admit no canonical realization**. Thus the failure of finite-grain separation is not automatically repaired by passing to the completion. The obstruction may persist in a different form, namely as failure of canonical branch selection at the inverse-limit level.

The paper is therefore organized around two obstructions:

1. **finite-grain blindness**: no monotone separator determined at any bounded level of the filtration can resolve the P–NP boundary;
2. **completion-level non-canonicity**: even after assembling all finite levels into the inverse-limit object, persistent frontier structure need not admit canonical realization.

The point of the paper is not to prove $P \neq NP$, nor to argue for independence. Its contribution is to sharpen the structural description of the obstruction. Any successful separator must be essentially asymptotic and must also overcome the possibility that the full observational tower still fails to determine a canonical computational branch.

The paper is organized as follows:

- Section 2 defines the computational field, the finite-grain filtration, and the topology of finite observability,
- Section 3 proves the finite-grain blindness theorem,
- Section 4 constructs the inverse-limit completion,
- Section 5 introduces realization towers and canonical realization,
- Section 6 proves the no-go theorem for canonical realization on persistent non-rigid frontier points,
- Section 7 applies this framework to boundary-induced completion profiles arising from the P–NP boundary,
- Section 8 discusses methodological consequences and open directions.

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2. The Computational Field

We work in a deliberately minimal setting. The purpose of this section is to define the ambient space, the finite-grain filtration, and the topology induced by finite observability, without making any assumption about the truth or falsity of $P = NP$. The perspective is structural: the space of decision problems is treated as a field of computational possibilities equipped with progressively finer forms of observation.

2.1. The space of languages

Let

$$X := \mathcal{P}(\{0,1\}^*)$$

be the class of all languages over the binary alphabet. Elements of X are denoted by $A, B, L, \ldots$. Throughout the paper, languages are treated extensionally, independently of any particular machine model or encoding beyond their identity as subsets of $\{0,1\}^*$.

The choice of X is canonical for the present purpose: the P vs NP problem concerns classes of decision problems, and therefore naturally lives in the ambient space of all languages.

2.2. Polynomial-time reducibility

We fix polynomial-time many-one reducibility as the basic comparison relation on X [**5**, **6**]. For $A, B \in X$, we write

$$A \leq_p B$$

if there exists a polynomial-time computable function $f: \{0,1\}^* \to \{0,1\}^*$such that

$$x \in A \iff f(x) \in B \text{ for all } x \in \{0,1\}^*.$$

We denote by

$$A \equiv_p B$$

the induced equivalence relation:

$$A \equiv_p B \iff A \leq_p B \text{ and } B \leq_p A.$$

For $A \in X$, define the associated reducibility cones:

$$\downarrow A := \{B \in X : B \leq_p A\}, \uparrow A := \{B \in X : A \leq_p B\}.$$

These sets encode the local geometry induced by reducibility: $\downarrow A$ consists of languages no harder than A, while $\uparrow A$ consists of languages at least as hard as A. They will later appear as topologically stable regions.

2.3. Finite-grain filtration

The central structural ingredient is a filtration of finite-grain computational indistinguishability. Intuitively, the index k measures how much computational structure is visible at resolution k, while the passage from k to $k+1$corresponds to refinement of observational granularity.

We fix a hierarchy of classes of total transformations

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$$

satisfying the following conditions:

1. each $\mathcal{F}_k$ consists of total functions
$$f : \{0,1\}^* \to \{0,1\}^*;$$
2. each $\mathcal{F}_k$ contains the identity map;
3. each $\mathcal{F}_k$ is closed under composition;
4. for every k, every $f \in \mathcal{F}_k$ is polynomial-time computable.

The intended meaning is that $\mathcal{F}_k$ captures the computational transformations observable at finite structural level k. No particular concrete choice of the filtration is needed for the general theory; only the formal properties above are used.

2.4. Finite-grain reducibility and indistinguishability

For $A, B \in X$, define

$$A \leq_k B$$

if there exists $f \in \mathcal{F}_k$ such that

$$x \in A \iff f(x) \in B \text{ for all } x.$$

We then define finite-grain indistinguishability at level k by

$$A \sim_k B \iff (A \leq_k B \text{ and } B \leq_k A).$$

For each k, $\leq_k$ is a preorder and $\sim_k$ is an equivalence relation. Moreover, refinement is monotone:

$$\ell \geq k \Rightarrow A \sim_\ell B \implies A \sim_k B.$$

Thus the family $(\sim_k)$ forms a genuine filtration of indistinguishability relations.

For each k, let

$$X_k := X/\sim_k$$

be the corresponding quotient space. If $\ell \geq k$, the monotonicity of the filtration induces a natural forgetfulness map

$$r_{\ell k} : X_\ell \to X_k$$

sending a finer class to the coarser class containing it.

This produces an inverse system

$$X_0 \overset{r_{10}}{\leftarrow} X_1 \overset{r_{21}}{\leftarrow} X_2 \leftarrow \cdots$$

which records computational observability at all finite levels simultaneously.

2.5. Neighborhoods and finite observability

For $A \in X$ and $k \in \mathbb{N}$, define the k-neighborhood of A by

$$N_k(A) := [A]_{\sim_k}$$

These neighborhoods represent all languages indistinguishable from A at finite granularity k. By monotonicity of the filtration,

$$N_{k+1}(A) \subseteq N_k(A)$$

for every k. Thus the finite-grain neighborhoods become progressively finer as the observational scale increases.

The intended interpretation is straightforward: if $B \in N_k(A)$, then A and B cannot be separated by any observation expressible through the structural resolution encoded by $\mathcal{F}_k$.

2.6. The topology of finite observability

The neighborhoods $N_k(A)$ induce a topology on X. Define

$$\tau := \{U \subseteq X : \forall A \in U, \exists k \text{ such that } N_k(A) \subseteq U\}.$$

This topology formalizes finite observability:

1. open sets correspond to properties decidable at some finite structural granularity around each point;
2. topological indistinguishability reflects irreducible finite-grain symmetry;
3. convergence corresponds to stabilization under increasingly fine observation.

The topology τ is generally non-Hausdorff. This is not a defect but part of the intended structure: languages that cannot be separated at any finite level of the filtration should not be separated topologically.

2.7. Stable classes

A class $\mathcal{C} \subseteq X$ is said to be downward closed under $\leq_p$ if

$$A \in \mathcal{C}, B \leq_p A \implies B \in \mathcal{C}.$$

The relevance of this notion is that every downward-closed class is open in (X, τ). In particular, both P and NP are downward closed under polynomial-time many-one reducibility, and therefore form open regions of the computational field [**3**, **5**, **6**].

This observation is fundamental for the geometric reading of the P vs NP problem developed later: the difficulty is not located inside P or inside NP, but on the boundary structure between them.

2.8. Finite levels and their completion

The finite-grain quotients X_k do not exhaust the structure encoded by the filtration. What they provide is a tower of partial observations. The associated completion is the inverse limit [**7**]

$$\hat{X} := \lim_{k} \leftarrow \; X_k$$

that is, the set of coherent tuples

$$a = (a_0, a_1, a_2, \dots), a_k \in X_k$$

such that for all $\ell \geq k$,

$$r_{\ell k}(a_\ell) = a_k$$

This completion is the natural limit object of the finite observational tower [**4**, **7**]: it records all compatible finite-grain views simultaneously. It plays, in the present setting, the same structural role that completion plays in more classical observational frameworks: it is not a new finite level, but the object in which the full compatibility pattern of all finite levels is gathered.

Each language $L \in X$ defines a canonical point of the completion by

$$\iota(L) := ([L]_{\sim_0}, [L]_{\sim_1}, [L]_{\sim_2}, \dots) \in \hat{X}.$$

The map

$$\iota: X \to \hat{X}$$

embeds each language into the completion via its entire observational profile across the filtration.

2.9. Structural interpretation

The computational field introduced here has two distinct but related layers.

At the finite-grain level, one studies the topology τ induced by the neighborhoods $N_k(A)$, and asks which properties are visible at bounded structural resolution. This is the level at which finite-grain blindness arises.

At the limit level, one studies the completion $\hat{X}$, where the issue is no longer mere finite visibility, but the organization of all finite observations into coherent inverse-limit data. The question there is not simply whether a point is realizable, but whether it is realizable canonically.

This distinction between finite observability and limit compatibility will be central in the rest of the paper.

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3. Stable Regions and Finite-Grain Blindness

Having introduced the computational field (X, τ), we now study how standard complexity-theoretic classes sit inside the topology induced by finite observability. The first point is that the basic regions determined by polynomial-time reducibility are topologically stable. The second, and more important, point is that the P–NP difficulty is concentrated on a boundary region that cannot be resolved by monotone finite-grain criteria. This is the first main obstruction of the paper.

3.1. Reducibility cones as stable regions

For a language $A \in X$, recall the reducibility cones

$$\downarrow A := \{B \in X : B \leq_p A\}, \uparrow A := \{B \in X : A \leq_p B\}.$$

These sets define the basic monotone regions of the computational field. Their topological significance is immediate: both lower and upper cones are open in the topology of finite observability. Intuitively, this means that reducibility generates locally stable regions in which finite-grain perturbations do not destroy hardness relationships. In particular, reducibility cones behave as genuine open neighborhoods of computational behavior, rather than merely order-theoretic shadows.

More generally, any class $\mathcal{C} \subseteq X$ that is downward closed under polynomial-time reducibility is open in (X, τ). Since both P and NP are downward closed, it follows that they form open subsets of the computational field. This is the first geometric manifestation of the framework: P and NP are not treated merely as syntactic classes, but as topologically stable regions of the ambient space of languages [**5**].

3.2. The boundary of polynomial-time solvability inside NP

Because P is open, its difficulty is not located in its interior. The relevant region is its boundary inside NP. Let

$$\partial P$$

denote the topological boundary of Pin (X, τ). Since $P \subseteq NP$ and NP is open, the set

$$\partial P \cap NP$$

naturally represents the frontier of polynomial-time solvability inside NP. This is the region where finite-grain perturbations no longer decide on which side of the inclusion a problem lies.

A key characterization is neighborhood-theoretic: a language $A \in NP$ lies on the boundary of P if and only if $A \notin P$ and every finite-grain neighborhood $N_k(A)$ intersects P. Equivalently, points on the frontier are arbitrarily close, at every finite level of observation, to polynomial-time solvable languages. The separation problem is therefore not a matter of isolated hard points; it is a matter of a boundary region that remains locally entangled with P at every finite granularity.

This yields an immediate topological reformulation of the P vs NP question: $P = NP$ if and only if the frontier is empty, and equivalently if and only if P is closed inside NP. In this sense, the classical complexity question becomes a geometric one about the existence and structure of a boundary.

3.3. Monotone invariants and local constancy

We now consider functions

$$\Phi: X \to Y$$

valued in a preordered set Y, and assume that Φ is monotone with respect to polynomial-time reducibility:

$$A \leq_p B \implies \Phi(A) \preceq \Phi(B).$$

Such a Φ represents the abstract form of a hardness-respecting invariant or criterion. The crucial observation is that finite-grain indistinguishability forces local constancy of every such invariant. If

$$A \sim_k B,$$

then $A \leq_k B$ and $B \leq_k A$, hence in particular $A \leq_p B$ and $B \leq_p A$. By monotonicity, $\Phi(A) = \Phi(B)$. Therefore every monotone invariant is constant on each finite-grain equivalence class and thus locally constant on the neighborhood basis $\{N_k(A)\}$. In particular, every monotone invariant is automatically continuous with respect to the topology τ.

This is the core local rigidity phenomenon: any invariant that respects the standard semantics of computational hardness is topologically flat on each finite neighborhood. The topology has been

engineered precisely so that finite-grain observational indistinguishability becomes local constancy for monotone criteria.

3.4. Boundary blindness

The local constancy result has a decisive consequence for the frontier. Let

$$A \in \partial P \cap NP.$$

By the neighborhood characterization, every finite neighborhood $N_k(A)$ meets both P and $NP \setminus P$. But every monotone invariant is constant on $N_k(A)$. Hence no such invariant can locally distinguish the side of the boundary on which a point lies. This yields the first main no-go statement:

Boundary blindness.
No monotone invariant can locally distinguish points on the P–NP boundary.

In other words, every hardness-respecting finite-grain criterion is intrinsically blind precisely where the separation problem lives. The obstruction is not a matter of poor choice of invariant, codomain, or implementation. It is built into the topology of finite observability itself.

3.5. Global potentials remain finite-grain blind

The previous argument is local. One might hope that a globally defined monotone potential could overcome local blindness by aggregating information across neighborhoods. The framework shows that this hope fails whenever the potential is determined at some finite level of the filtration.

A global monotone potential

$$\Phi: X \to Y$$

is said to be finite-grain determined if it factors through some quotient $X_k = X/{\sim_k}$. Equivalently, there exists a finite level k such that Φ depends only on the class of a language modulo $\sim_k$. But then Φ is constant on $N_k(A)$, and therefore cannot distinguish points of P from points of $NP \setminus P$ on the boundary. Thus globality does not help as long as the functional compresses the infinite filtration to a finite stage.

This strengthens the obstruction:

1. monotonicity implies local constancy on all finite neighborhoods;
2. finite-grain determination implies global blindness on the boundary.

Hence the issue is not locality per se, but finite granularity. Even a globally defined hardness-respecting potential fails if it collapses the filtration at any bounded level.

3.6. Structural consequence

The conclusion of this section is structural rather than technical. The failure mechanism identified here is not specifically combinatorial, algebraic, or proof-theoretic. It is topological. Finite observability induces a topology on the computational field; P and NP become open stable regions; and the difficulty of the separation problem is concentrated on their boundary. Every monotone finite-grain criterion is flat on finite neighborhoods and therefore blind precisely where the problem lives.

This does not establish $P \neq NP$, nor does it prove independence. What it does establish is a sharp no-go theorem for a broad family of approaches: any method based on hardness-respecting measures, finite-level invariants, or bounded structural compression is incapable of resolving the boundary. Consequently, any successful separator must depend on the **entire filtration**, not on any bounded stage of it. In the language introduced later in the paper, any viable separator must be essentially asymptotic [**1**, **2**, **8**].

3.7. Transition to the completion

Section 3 identifies the first obstruction: finite-grain blindness on the boundary. The natural next question is whether passing to the inverse-limit completion of the filtration removes that obstruction. The completion records all compatible finite observations simultaneously, and is therefore the most natural candidate for a limit object in which the residual difficulty might be studied. The next sections show that even there, the problem does not automatically disappear: the remaining obstruction may be one of **canonical realization**, not merely one of finite visibility.

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4. The Inverse-Limit Completion of the Filtration

Let

$$X := \mathcal{P}(\{0,1\}^*)$$

and let

$$(\sim_k)_{k\in\mathbb{N}}$$

be the finite-grain filtration introduced in Section 2 **[4, 7]**.

For each k, define

$$X_k := X/\sim_k$$

Since the filtration is monotone, for every $\ell \geq k$ there is a well-defined forgetfulness map

$$r_{\ell k} : X_\ell \to X_k, r_{\ell k}([A]_{\sim_\ell}) = [A]_{\sim_k}$$

Definition 4.1

The **completion** of the filtration $(\sim_k)$ is the inverse limit [**7**]

$$\hat{X} := \lim_{k} \leftarrow X_k$$

that is,

$$\hat{X} = \left\{ (a_k)_{k\in\mathbb{N}} \in \prod_k X_k : r_{\ell k}(a_\ell) = a_k \text{ for all } \ell \geq k \right\}$$

Definition 4.2

For each $A \in X$, define

$$\iota(A) := ([A]_{\sim_0}, [A]_{\sim_1}, [A]_{\sim_2}, \dots) \in \hat{X}$$

Proposition 4.3

The map

$$\iota : X \to \hat{X}$$

is well-defined.

Proof

For every $A \in X$ and every $\ell \geq k$,

$$r_{\ell k}([A]_{\sim_\ell}) = [A]_{\sim_k}.$$

Hence the tuple $\iota(A)$ satisfies the coherence conditions defining $\hat{X}$. ■

Remark 4.4

The object $\hat{X}$is not a new finite level of observation. It is the unique object encoding all compatible finite-grain observations simultaneously.

Definition 4.5

For $a = (a_k)_{k \in \mathbb{N}} \in \hat{X}$, define its k-th realization set by

$$\mathrm{Rep}_k(a) := \{A \in X : [A]_{\sim_k} = a_k\}.$$

Thus a point $a \in \hat{X}$ determines, level by level, a family of admissible representatives in X.

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5. Realization Towers and Canonical Realization

The completion $\hat{X}$ records inverse-limit data **[7]**. The next question is whether such data admit coherent realizations in X, and if so, whether such realizations are canonical.

Definition 5.1

Let $a = (a_k)_{k \in \mathbb{N}} \in \hat{X}$.
A **realization tower** of a is a sequence

$$(A_k)_{k \in \mathbb{N}}$$

such that:

$$A_k \in \mathrm{Rep}_k(a) \text{ for all } k,$$

and

$$[A_\ell]_{\sim_k} = [A_k]_{\sim_k} \text{for all } \ell \geq k.$$

Equivalently, a realization tower is a coherent branch through the family

$$\mathrm{Rep}_0(a) \leftarrow \mathrm{Rep}_1(a) \leftarrow \mathrm{Rep}_2(a) \leftarrow \cdots .$$

Definition 5.2

An **internal selection rule** for $a \in \hat{X}$ is a rule assigning to each k an element

$$s_k(a) \in \mathrm{Rep}_k(a).$$

Definition 5.3

An internal selection rule $s = (s_k)_k$is called:

- **coherent** if
$$[s_\ell(a)]_{\sim_k} = [s_k(a)]_{\sim_k} (\ell \geq k);$$
- **internally definable** if each $s_k(a)$ depends only on the truncated inverse-system data
$$(a_0, \dots, a_k)$$
and the maps $r_{\ell k}$, and not on external representatives or auxiliary conventions;
- **refinement-stable** if the choice at level k is preserved under passage to level $k + 1$.

Definition 5.4

A **canonical realization** of $a \in \hat{X}$ is an internal selection rule satisfying the three conditions above.

Definition 5.5

A point $a \in \hat{X}$ is **internally rigid** if it admits exactly one canonical realization.

It is **internally non-rigid** if:

- either no canonical realization exists,
- or more than one canonical realization exists.

Proposition 5.6

If $a \in \hat{X}$ is internally rigid, then the inverse-limit data of a determine a unique structurally privileged branch through the realization system.

Proof

Immediate from Definition 5.5. ■

Remark 5.7

The distinction between **realizability** and **canonical realizability** is essential. A point of $\hat{X}$ may admit coherent realization towers without determining any unique privileged one.

6. Persistent Frontier Structure and Canonical Obstruction

We now define the class of limit points whose ambiguity is not removed at any finite stage.

Definition 6.1

A point

$$a = (a_k)_{k\in\mathbb{N}} \in \hat{X}$$

is a **persistent frontier point** if, for every k, the component $a_k \in X_k$ remains unresolved at level k, and this unresolved character is not eliminated by any finite refinement.

Definition 6.2

A point $a \in \hat{X}$ is an **internal persistent frontier point** if, for every k, no rule definable from the data

$$(a_0, \dots, a_k)$$

canonically determines a unique representative behavior compatible with all finer levels.

Remark 6.3

Definition 6.2 is the internal completion-theoretic version of the frontier notion: the ambiguity is not accidental and does not disappear at any finite observational scale.

Theorem 6.4

Let $a \in \hat{X}$. If a is an internal persistent frontier point and is internally non-rigid, then a admits no canonical realization.

Proof

Assume, toward contradiction, that $a$admits a canonical realization $s = (s_k)_k$.

By definition, s is:

1. coherent,
2. internally definable,
3. refinement-stable.

Hence s determines a structurally privileged branch through the realization system

$$\mathrm{Rep}_0(a) \leftarrow \mathrm{Rep}_1(a) \leftarrow \mathrm{Rep}_2(a) \leftarrow \cdots$$

using only internal data of the inverse tower.

But a is an internal persistent frontier point, so no finite truncation canonically resolves the ambiguity encoded by a. Moreover, a is internally non-rigid, so the full inverse-limit data do not determine a unique privileged branch.

Therefore the assumed canonical realization must fail in one of two ways:

- either it is not unique, contradicting canonicity;
- or it depends on extra structure not present in the inverse system, contradicting internal definability.

Both are impossible. Hence no canonical realization exists. ■

Corollary 6.5

If $a \in \hat{X}$ is an internal persistent frontier point and admits a canonical realization, then a must be internally rigid.

Proof

Contrapositive of Theorem 6.4. ■

Definition 6.6

Let

$$F \subseteq \hat{X}.$$

We say that F is a **persistent frontier region** if every $a \in F$ is an internal persistent frontier point.

Proposition 6.7

If $F \subseteq \hat{X}$ is a persistent frontier region and every point of F is internally non-rigid, then no point of F admits canonical realization.

Proof

Apply Theorem 6.4 pointwise. ■

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7. Computational Boundary and Completion

We now connect the abstract completion-theoretic notions of Sections 4–6 with the computational boundary introduced in Section 3.

Definition 7.1

For $A \in X$, define its completion profile by

$$\iota(A) := ([A]_{\sim_0}, [A]_{\sim_1}, [A]_{\sim_2}, \dots) \in \hat{X}.$$

Proposition 7.2

Let

$$A \in \partial P \cap NP.$$

Then, for every k,

$$[A]_{\sim_k} \cap P \neq \emptyset.$$

Proof

By the boundary characterization established in the finite-grain topology, every neighborhood $N_k(A)$ intersects P. Since

$$N_k(A) = [A]_{\sim_k}$$

the claim follows. ■

Proposition 7.3

Let

$$A \in \partial P \cap NP.$$

Then, for every k, the component

$$[A]_{\sim_k} \in X_k$$

is unresolved with respect to polynomial-time solvability.

Proof

By Proposition 7.2, the class $[A]_{\sim_k}$contains at least one language in P. Since $A \in NP \setminus P$, the same class also contains a language outside P, namely A itself. Hence $[A]_{\sim_k}$ does not determine polynomial-time solvability at level k. ■

Corollary 7.4

Let

$$A \in \partial P \cap NP.$$

Then $\iota(A) \in \hat{X}$ is a persistent frontier candidate.

Proof

By Proposition 7.3, every finite component of $\iota(A)$ is unresolved. Therefore no finite stage of the inverse tower canonically collapses the ambiguity encoded by $\iota(A)$. ■

Definition 7.5

A point

$$a \in \hat{X}$$

is called a **boundary-induced completion profile** if there exists

$$A \in \partial P \cap NP$$

such that

$$a = \iota(A).$$

Proposition 7.6

Every boundary-induced completion profile is a persistent frontier candidate.

Proof

Immediate from Corollary 7.4. ■

Proposition 7.7

Let

$$A \in \partial P \cap NP.$$

If $\iota(A)$ is internally non-rigid, then $\iota(A)$ admits no canonical realization.

Proof

By Proposition 7.6, $\iota(A)$ is a persistent frontier candidate. If $\iota(A)$ is moreover internally non-rigid, then Theorem 6.4 applies. ■

Remark 7.8

Proposition 7.7 is conditional. It does not assert that every boundary-induced completion profile is internally non-rigid. It states only that internal non-rigidity, when present, obstructs canonical realization.

Proposition 7.9

Let

$$A \in \partial P \cap NP.$$

Then no finite-grain monotone invariant canonically resolves the completion profile $\iota(A)$ at any finite level.

Proof

For every k, the component $[A]_{\sim_k}$is unresolved by Proposition 7.3. On the other hand, every monotone finite-grain invariant is constant on $[A]_{\sim_k}$, by the local constancy result of Section 3. Hence no such invariant distinguishes the polynomial-time side at level k. ■

Theorem 7.10

Let

$$A \in \partial P \cap NP.$$

Then the completion profile $\iota(A)$ cannot be canonically resolved by any rule determined at a finite stage of the filtration.

Proof

Suppose such a rule existed. Then there would exist $k$such that the canonical resolution of $\iota(A)$ depended only on the finite component $[A]_{\sim_k}$. But $[A]_{\sim_k}$ is unresolved by Proposition 7.3, and finite-grain monotone criteria are constant on this class by Proposition 7.9. Therefore the rule cannot distinguish the relevant side of the boundary. Contradiction. ■

Corollary 7.11

If a boundary-induced completion profile admits canonical realization, such realization cannot be finite-grain determined.

Proof

Immediate from Theorem 7.10. ■

Structural consequence

Sections 3 and 7 together yield the following dichotomy for boundary-induced completion profiles:

1. finite-grain criteria do not resolve them at any bounded level;
2. if they are internally non-rigid, then they admit no canonical realization at all.

Thus the completion does not automatically trivialize the computational boundary. It transforms a finite-level blindness phenomenon into a limit-level question of canonicity.

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8. Conclusion and Open Directions

We summarize the structural content of the paper.

First, the finite-grain filtration

$$(\sim_k)_{k\in\mathbb{N}}$$

induces a topology on

$$X = \mathcal{P}(\{0,1\}^*)$$

in which $P$and $NP$are open regions and the computational difficulty is concentrated on the boundary

$$\partial P \cap NP.$$

Within this topology, every monotone separator determined at a finite stage of the filtration is locally constant and therefore blind on the boundary. This is the first obstruction: **finite-grain blindness [1, 2, 8]**.

Second, the inverse system of finite-grain quotients gives rise to the completion

$$\hat{X} = \lim_{k} \leftarrow \; X/{\sim_k}.$$

This is the natural limit object associated with the filtration **[4, 7]**. However, the passage to $\hat{X}$ does not automatically restore canonical structure. The notions of realization tower and canonical realization show that existence of coherent inverse-limit data is weaker than existence of a canonically determined branch. The second obstruction is therefore **completion-level non-canonicity**: persistent non-rigid frontier points do not admit canonical realization.

Taken together, the results of the paper yield the following picture. A successful separator for P versus NP, if one exists, cannot be finite-grain determined. But neither is it enough to pass formally to the inverse-limit completion of the filtration. One must also overcome the possibility that the completion fails to determine a canonical computational branch. Thus the obstruction is twofold:

$$\text{finite-grain blindness} + \text{failure of canonical realization in the completion.}$$

The paper leaves open several questions.

Open Problem 1

Determine whether every boundary-induced completion profile

$$\iota(A), A \in \partial P \cap NP,$$

is internally non-rigid.

A positive answer would strengthen the completion-level obstruction substantially.

Open Problem 2

Characterize those filtrations $(\sim_k)$ for which the completion $\hat{X}$ admits canonical realization on a nontrivial class of points.

This question is independent of P versus NP and belongs to the general theory of inverse-limit observational systems.

Open Problem 3

Define and study canonical residual structure in the completion, for example through subsets of the form

$$R_{\text{can}} := \widehat{NP} \setminus \text{Norm}_P^{\text{can}},$$

and determine under what conditions such residual sets are nonempty.

This would isolate, at the limit level, the part of NP not canonically absorbable into P.

The main contribution of the paper is therefore not a resolution of P vs NP, but a refinement of its structural formulation. The finite-grain viewpoint identifies the boundary as the locus of unavoidable blindness. The completion viewpoint shows that even the natural limit object of the filtration may fail to canonically resolve that boundary. Any future approach must address both phenomena

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